

Department of Computer Science and Engineering
Academic Year 2020– 2021 (Odd Semester)

Degree, Semester & Branch: B.E, V, CSE

Course Code & Title: CS501 & Theory of Computation

Name of the Faculty member (s): Dr.I.Gethzi Ahila Poornima

Innovative Practice Description

- **Unit / Topic: Unit V / Undecidable Problems**

- **Course Outcome: CO5**

- **Topic Learning Outcome: TLO2**

- **Activity Chosen: Flipped Class room**

- **Justification:**

Flipped classroom is a pedagogical approach in which traditional content delivery moves to an individual self-learning process, and the group activity is transformed into a dynamic, interactive learning environment where the instructor just guides the learners to apply their creative thinking to provide a fruitful solution in the course content. It provides more benefits than traditional direct content delivery.

This topic undecidable problem is somewhat vague concept which needs more attention to make the students understood. By doing group discussion, the concept become more familiar to the students.

- **Time Allotted for the Activity: 50 Minutes**

- **Details of the Implementation:**

Based on the reason given in justification, the topic is chosen and announced in the class four days prior. The students are asked to refer the NPTEL video given in the classroom already and come prepared for the group discussion. Also, they are instructed to take notes accordingly. On the day of activity, the small groups are formed and made them to discuss with the peer team members. A few sample snaps taken during the event are shown in Figure 1.

- **CO – PO / PSO mapping:**

CO	PO2	PO4
CO2	1	2

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO2	PO4
	1	2
Justification for	Students could identify modules or algorithms for	• Students will be able to

correlation	solution of engineering problems	investigate a system based on their ability to solve problems.
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• **Images / Screenshot of the practice:**



Figure 1: Screenshots during the activity

• **Reflective Critique:**

❖ ***Feedback of practice from students and other stakeholders:***

- The students gave a very good feedback about this activity. They felt that, they could be more thorough with this concept by doing this activity.

❖ ***Benefit of the practice:***

- Learning at home automatically becomes student-centric rather than teacher-centric.
- They observed and learned a few things individually that lead to self-learning. They actively involved in this session rather than it being a one-way communication.

❖ ***Challenges faced in implementation:***

- Deadlock situation may happen during an in-class activity like brainstorming.
- A few students were not capable of understanding the concept, but they overcome by discussing with peer team members.

Signature of Faculty Member

HOD